

Lab - 1 : Set - 1 : The Logistic Modelling of IBM

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In this lab, we analytically analyze and model **IBM's** human resources, revenue, and profit over time using the logistic equation. We then compare these data points with the theoretical model by calculating the mean and standard deviation to assess their accuracy. Additionally, we plot the correlated growth of human resources and revenue to examine their relationship.

I. MODEL

First-order autonomous dynamical systems have the general form of

$$\frac{dx}{dt} = f(x) \quad (1)$$

Eq. (1) where $x = x(t)$, with t being time. This system may be linear or non-linear, depending on whether $f(x)$ is, respectively, a linear or non-linear function of x . A basic model of a non-linear function is given by

$$f(x) = ax - bx^2 \quad (2)$$

with a and b being fixed parameters. This leads to the well-known logistic equation,

$$\frac{dx}{dt} = f(x) = ax - bx^2 \quad (3)$$

Under the initial condition of $x(0) = x_0$, and with the definition of $k = a/b$, the integral solution of Eq. (1) is

$$x(t) = \frac{kx_0e^{at}}{k + x_0(e^{at} - 1)} \quad (4)$$

The non-linear time scale is given as,

$$t_{nl} = \frac{1}{a} \ln\left(\frac{k}{x_0} - 1\right) \quad (5)$$

The following formula gives the linearisation of human resources and revenue:

$$V \sim U^\beta \quad (6)$$

II. RESULTS

• Annual revenue growth plot of IBM from 1914 to 2006

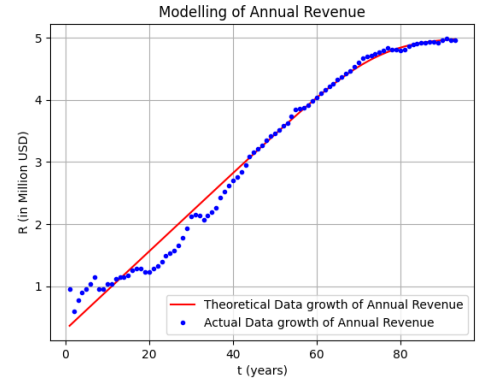


FIG. 1: Shows the net revenue growth of IBM company over 75-80 years. The parameter values used to fit the revenue growth are $\rho_1 = 0.145 \text{ year}^{-1}$ and $k_R = 100 \text{ billion USD}$, For the logistic model, the yearly relative variation in revenue has a mean $\mu_R = 0.025$ and a standard deviation $\sigma_R = 0.4897$.

• Human resources growth of IBM from 1914 to 2006

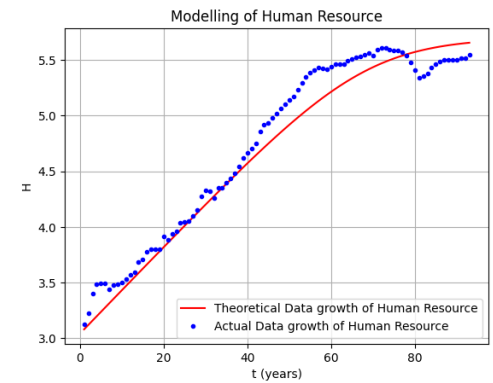


FIG. 2: Shows the net human resource growth of IBM company over 75-80 years. The parameter values used to fit the revenue growth are $\eta_1 = 0.09 \text{ year}^{-1}$ and $k_H = 500000$, For the logistic model, the yearly relative variation in revenue has a mean $\mu_H = 0.106$ and a standard deviation $\sigma_H = 0.2999$.

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• Net annual Profits of IBM from 1914 to 2006

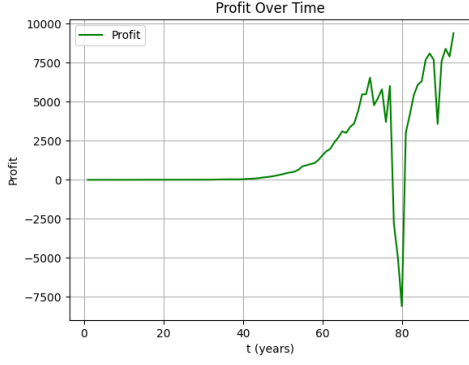


FIG. 3: The net annual profits of IBM grow steadily till about 75-80 years. Around this time IBM suffered major losses in its net earnings (\$ 8 billion in 1993), and this time scale corresponds closely to the time scale for the onset of nonlinear saturation in revenue growth.

• Correlated growth of annual revenue and human resource strength of IBM

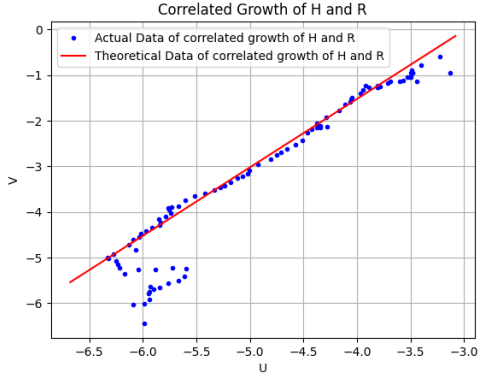


FIG. 4: Shows the correlation between Human resources and Net revenue of IBM over the timescale of 75-80 years. For the dotted graph, $U = H^{-1} - k_H^{-1}$ and $V = R^{-1} - k_R^{-1}$. For the line graph, $U = H^{-1} - k_H^{-1}$ and $V = \alpha \cdot U^\beta$, where $\beta = \frac{\rho_1}{\eta_1} \approx 1.5$, $\alpha = 30000$, $k_H = 500000$, and $k_R = 100000$. The log-log fits the power law very well except for the bottom left.

III. STATISTICS

Category	Mean	Standard Deviation
Annual Revenue	0.024998404862919595	0.48704047134413464
Human Resource	0.10545862329253444	0.30326598309485664

IV. CONCLUSIONS

- Figures 1 and 2 show that the modeled curve follows the overall trend of annual revenue and human resource. Over time, it saturates at a constant value, meaning growth slows down and stabilizes.
- Figures 2 and 3 indicate a major drop in IBM's profits in 1993, which occurs at the same time as a decline in human resource growth. This suggests that financial struggles also affected the workforce.
- Figure 4 shows that the power law model matches the actual data in the log-log plot, confirming a strong link between H and R. This means that when one value changes, the other tends to follow a similar pattern, supporting the accuracy of the model.

[1] Arnab K. Ray, *Logistic modelling of economic dynamics*, DA-IICT, (September 6, 2023).